

DESIGN & SETUP REPORT

A transcranial focused-ultrasound targeting pipeline for the squirrel monkey: a *Saimiri sciureus* museum microCT registered to the VALiDATE29 brain atlas, with a hard CT-calibration guard (deliverables 1–2)

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Abstract. We document the fifth pillar of TUBA — a transcranial focused ultrasound (FUS) pipeline for the common squirrel monkey (*Saimiri sciureus*), prepared for NIH R01 EB037345 (Aim 3): per-animal skull models for time-reversal aberration correction and multifocal beam design targeting the hand representations of primary somatosensory (S1 areas 3b and 1) and primary motor (M1, area 4) cortex. This report covers **deliverables 1–2 (the geometry stages)** and is scoped, by design decision, to a *geometry-only* build: no HU-calibrated colony CT is available yet, so the pillar is bootstrapped on an *uncalibrated* museum microCT (*Saimiri sciureus* USNM 338948, UT Austin / DigiMorph, 0.1189 mm isotropic) registered to the VALiDATE29 multi-channel MRI atlas of the squirrel-monkey brain (Schilling *et al.* 2017; 29 animals, histologically-defined cortical labels). The pillar reuses TUBA's core primitives from the mouse/rat/macaque pillars almost unchanged — bone-shell cavity extraction (`tuba.core.cavity.bone_shell_cavity`), `cavity_binary` ANTs SyN (affine + deformable), atlas-into-subject warp, and the beam-aligned (c, ρ) slab export. Its *distinctive* contribution is the **CT-calibration guard** for deliverable 1: a new module `tuba.core.hu_acoustics` implements the Aubry (2003) porosity $HU \rightarrow (\rho, c, \alpha)$ model but *refuses to run* on input not flagged Hounsfield-calibrated (`UncalibratedInputError`), so a museum microCT can never silently produce quantitatively wrong acoustics; an explicitly-flagged placeholder ramp is the sanctioned geometry-only fallback. Deliverable 1 also reports the points-per-wavelength (PPW) budget across the 2–4 MHz operating band on the simulation grid, and a grid-convergence run of the transcranial time-of-flight aberration (the scalar a time-reversal correction must undo). All constants that can only be fixed against the *staged* scan (orientation flips, intensity thresholds, cavity hull geometry) are marked `PROVISIONAL`; quantitative cavity/registration results and QC figures are produced by the end-to-end driver once the scan is staged. The downstream wave physics (deliverables 3–7: time-reversal, multifocal synthesis, the two-foci feasibility sweep, MI/bioheat safety, and the per-animal export artifact) is out of scope for this branch and consumes the (c, ρ, dz) slab in the `fullwave2-*` solver siblings. Code: `src/tuba/species/saimiri.py`, `tuba.core.hu_acoustics`, `tuba.atlases.validate29`; data manifest + fetchers at `src/tuba/data/`.

1 Scope and status

This is a *design and setup* report, not a validation report: the four prior TUBA pillars (mouse, macaque, human, rat) each document a pipeline calibrated end-to-end against a staged scan with measured cavity volumes, centroids, and focal figures. The squirrel-monkey pillar is at an earlier stage by deliberate choice — the funded project targets a Vanderbilt colony, whose HU-calibrated CT is not yet in hand, so we build and verify the *machinery* on a public museum microCT for geometry, gating the acoustics behind a calibration guard (§4) so nothing downstream mistakes an uncalibrated skull for a quantitative one. Two decisions frame the branch:

1. **Subject data:** a public museum microCT (§2), geometry only. The $HU \rightarrow \text{acoustics}$ step refuses to run on it.
2. **Scope:** deliverables 1–2 (CT $\rightarrow$ acoustic model + atlas registration + target export + QC). Deliverables 3–7 (the wave-solver study) are out of scope here and live in the `fullwave2-*` siblings (§7).

Where a quantity can only be measured on the staged scan we write **[pending staged scan]** rather than a fabricated number.

2 Source data

Skull. The skull volume is a microCT scan of a *Saimiri sciureus* cranium, museum specimen USNM 338948 (National Museum of Natural History), scanned at the University of Texas High-Resolution X-ray CT Facility (UTCT) and distributed on DigiMorph [1]. The reconstruction is 540 axial slices at 0.1189 mm isotropic over a 54.0 mm field of reconstruction — a dry cranium, no soft tissue. The same openVertebrate / oUTCT specimens are mirrored on MorphoSource, the collection the TUBA macaque pillar [9] also draws from (Copes *et al.* 2016 [2]). Table 1 summarises the parameters.

Table 1. Saimiri sciureus USNM 338948 scan parameters (DigiMorph specimen page). The intensity axis is *uncalibrated* (no HU phantom in the field of view).

Specimen	<i>Saimiri sciureus</i> , USNM 338948 (NMNH)
Facility	UT Austin High-Resolution X-ray CT (UTCT)
Slices	540 axial
Voxel	0.1189 mm isotropic
Field of reconstruction	54.0 mm
Calvarium thickness	1.0–1.5 mm (~ 8 – 13 voxels)
Intensity	uncalibrated reconstruction counts

Uncalibrated intensity — the central caveat. Museum microCT carries no hydroxyapatite/HU phantom, so voxel values are arbitrary reconstruction counts, not Hounsfield units. Every prior TUBA pillar faces the same issue and handles it by anchoring a histogram-relative intensity $\rightarrow (c, \rho)$ ramp on the cortical-bone mode [8, 9]: robust to detector scale, but explicitly *not* quantitatively calibrated. For the squirrel monkey we make that limitation a hard boundary rather than a footnote (§4): the scan is flagged `UNCALIBRATED_MICROCT` and quantitative acoustics are refused.

Wavelength / PPW criterion. The neuromodulation operating point is $f_0 = 2$ MHz with a 2–4 MHz design range. The most restrictive wavelength is in the water column at 4 MHz: $\lambda_{\min} = c_{\text{water}}/f = 1500/4 \times 10^6 = 0.375$ mm. A finite-difference solve wants $\gtrsim 6$ points per wavelength there, so the simulation grid pitch is set to $\Delta x = 60 \mu\text{m}$ (`SIM_DX_M` in `tuba.species.saimiri`); the resulting PPW budget is Table 2. Note the 0.1189 mm microCT is *coarser* than the $60 \mu\text{m}$ sim grid, so the skull is up-sampled onto the sim grid by the slab sampler (§6) — the geometry is faithful; only the absolute acoustics are placeholder.

Table 2. Points-per-wavelength budget on the $\Delta x = 60 \mu\text{m}$ simulation grid, using the shortest (water) wavelength at each frequency (`tuba.core.hu_acoustics.report_ppw`). All three bands clear the 6-PPW floor.

Frequency	λ_{water} (mm)	PPW @ $60 \mu\text{m}$
2.0 MHz	0.750	12.50
3.0 MHz	0.500	8.33
4.0 MHz	0.375	6.25

3 VALiDATE29 reference atlas

The brain anatomy is provided by the VALiDATE29 atlas of the squirrel monkey brain [3], built from 29 animals at the Vanderbilt University Institute of Imaging Science and distributed CC BY via NITRC (RRID:SCR_015542). VALiDATE29 is multi-channel: anatomical templates (proton-density, T_1 , T_2^*), diffusion templates (FA, MD), ex-vivo templates, and — critically for this project — **histologically-defined cortical labels** co-registered from Nissl/myelin sections, plus tractography-defined white-matter labels. It supersedes the earlier single-subject squirrel-monkey MRI atlas [4].

Layout tolerance. The exact filenames inside `VALiDATE29.zip` drift across releases 1.0/1.1/1.2 (several template channels, more than one label volume), so the binding `tuba.atlases.validate29.VALiDATE29` does not hard-code them: it *resolves* the template, the cortical-label volume, and the label table by globbing the staged directory with keyword heuristics, and raises with the candidate list when resolution is ambiguous rather than guessing. Every path is overridable by constructor argument or environment variable (`VALIDATE29_DEST`, `VALIDATE29_URL`).

Target structures. The study targets the *hand* representations in S1 areas 3b and 1 and in M1 (area 4); S1 and M1 are only ~ 2 – 3 mm apart in the squirrel monkey, which is precisely why the downstream two-foci feasibility question (§7) is non-trivial. Because the atlas label spellings vary across releases, `validate29.resolve_label` matches *stable area-number substrings* (`'area 3b'`, `'area 1'`, `'area 4'`) rather than a full canonical name; the exact spelling is confirmed against the staged label table (§9). Localising the hand sub-representation *within* each area additionally requires VALiDATE29’s stereotaxic frame (a mediolateral/AP prior on the homunculus), applied once the atlas coordinate frame is confirmed; until then `target_world_mm` returns the whole-area centroid and says so.

4 Deliverable 1: CT $\rightarrow$ acoustic model, with a calibration guard

This is the pillar’s distinctive contribution and the reason a geometry-only build is safe. The module `tuba.core.hu_acoustics` provides a *quantitative* HU mapping and a hard guard around it.

Porosity model. For Hounsfield-calibrated input we use the Aubry (2003) [5] porosity model. Bone porosity is estimated from CT radiodensity,

$$\Phi(\text{HU}) = 1 - \text{clip}\left(\frac{\text{HU} - \text{HU}_{\text{water}}}{\text{HU}_{\text{bone}} - \text{HU}_{\text{water}}}, 0, 1\right), \quad (1)$$

with $\Phi = 1$ water/marrow and $\Phi = 0$ dense cortical bone, and the acoustic properties interpolate linearly in the bone fraction $(1 - \Phi)$:

$$\rho(\Phi) = \rho_{\text{water}} + (\rho_{\text{bone}} - \rho_{\text{water}})(1 - \Phi), \quad (2)$$

$$c(\Phi) = c_{\text{water}} + (c_{\text{bone}} - c_{\text{water}})(1 - \Phi), \quad (3)$$

with cortical endpoints $\rho_{\text{bone}} = 1900 \text{ kg/m}^3$, $c_{\text{bone}} = 2900 \text{ m/s}$ consistent with the rat/macaque pillars, and water $\rho = 1000 \text{ kg/m}^3$, $c = 1500 \text{ m/s}$. Attenuation is taken proportional to bone fraction between a soft-tissue floor (0.5 dB/cm/MHz) and a cortical-bone value (8 dB/cm/MHz) [6, 7], reported in dB/cm/MHz and applied by the downstream solver. SI units throughout.

The guard. `AubryHUMapping.properties()` raises `UncalibratedInputError` unless the bound `CTCalibration.kind == 'hu'`. The squirrel-monkey scan is a museum microCT with no phantom, so the pillar binds `CT_CALIBRATION = UNCALIBRATED_MICROCT` and the quantitative path is refused by construction. Rather than silently emitting physically meaningless densities and sound speeds from arbitrary reconstruction counts, the caller must either (a) supply an HU-calibrated colony CT — one line, `CT_CALIBRATION = CTRCalibration(kind='hu', hu_bone_max=<scan>)` — which unlocks the quantitative mapping automatically, or (b) opt in explicitly to `placeholder_ramp()`, an `intensity → (c, ρ)` ramp carrying nominal cortical endpoints and a `.placeholder = True` tag so downstream code can detect the uncalibrated origin. The slab loader (§6) uses the flagged placeholder ramp: geometry is faithful, absolute (c, ρ) are nominal.

PPW report + grid convergence. `acoustic_model_report()` prints the calibration state, the PPW table (Table 2) at 2/3/4 MHz on the sim grid, and — given a placement and a staged scan — one grid-convergence run of the transcranial *time-of-flight aberration*

$$\tau = \int \left(\frac{1}{c(x)} - \frac{1}{c_{\text{ref}}} \right) dx, \quad (4)$$

the phase-insertion delay that a time-reversal / phase-conjugation correction must undo, computed along the central beam ray at Δx and $\Delta x/2$ and reported as an absolute and relative change. This is a convergence of the *acoustic-property* model with voxel pitch, not of a full-wave solve (which lives in the solver siblings, §7); it is exactly the quantity the downstream aberration correction consumes, so it is the right model-side convergence to report here. The units and sign are checked in the offline test suite (homogeneous medium $\Rightarrow \tau = 0$; a faster bone slab $\Rightarrow \tau < 0$; a smooth profile converges as $\Delta x \rightarrow 0$).

5 Deliverable 2: geometry — cavity, registration, targets

The geometry stages reuse the rat/macaque dry-skull spine with no algorithmic change; only the species constants differ, and the ones that depend on the scan are `PROVISIONAL` until it is staged.

Ingest + orientation (PROVISIONAL). The DigiMorph TIFF stack carries no DICOM affine; as in the rat pillar [10] the storage→anatomy mapping is fixed by an orientation probe (per-slice bone-mass profile + mid-stack orthoslices). The defaults in `tuba.species.saimiri` (`SWAP_ROW_COL=True`, `VOXEL_SIGNS=(-1, -1, +1)`, storage (slice=*AP*, row=*DV*, col=*LR*)) follow the macaque/rat museum-CT convention and are *provisional* pending that probe. At 0.1189 mm the scan is already near the rat working-cache pitch, so a factor-1 reformat to RAS is used rather than a block-max downsample.

Endocranial cavity (PROVISIONAL). Squirrel-monkey whole-brain volume is $\sim 22 \text{ mL}$, so the cavity target is $\sim 20\text{--}28 \text{ mL}$; the many dry-cranial openings (orbits, foramen magnum) rule out the mouse-style pure-hysteresis extractor for the same reason they do in rat and macaque. We therefore use the species-agnostic bone-shell extractor `tuba.core.cavity.bone_shell_cavity` (per-column basicranium floor + per-row plugs + per-axial hull), with hull half-extents interpolated between the rat ($\sim 2 \text{ mL}$) and macaque ($\sim 90 \text{ mL}$) pillars (`HULL_X_HALF_MM=18`, etc.). The produced volume is checked against the $20\text{--}28 \text{ mL}$ target at run time and flagged if outside; the measured value is `[pending staged scan]`.

Registration — cavity_binary SyN (affine + deformable). Deliverable 2 calls for affine *and* deformable registration, so — unlike the macaque/rat pillars, which stop at a 12-DOF Affine — we use the full `cavity_binary` SyN path (`tuba.core.warp.register_subject_to_atlas`, fixed = VALiDATE29 brain mask, moving = subject cavity), the same convention the mouse and mini pillars use. The atlas template and cortical labels are then pulled back into subject space through the saved inverse transforms (`warp_atlas_into_subject`, linear for the template, nearest-neighbour for the labels). The brain mask is derived from the label volume (`label > 0`), not a template threshold — the same lesson the rat and mouse pillars record.

Target export + QC (deliverable 2 outputs). `export_targets()` writes S1-3b, S1-area-1, and M1 hand-representation coordinates in the subject simulation frame to `saimiri_targets.json`, alongside the transform prefix, and `qc_figure()` writes the three-panel orthoslice overlay (bone shell, warped VALiDATE29 brain mask, S1/M1 markers). The transform is saved under `$TUBA_SAIMIRI_REG_DIR` with prefix `saimiri_cavity_to_validate29`. The exported coordinates and the QC figure are produced by `build()` once the scan + atlas are staged; their values are **[pending staged scan]**.

6 Placement and slab export

`place_on_skull` computes the apex/target/beam triple for a focused bowl aimed at a study target, and `load_slab` returns $(c, \rho, dz_{\text{slab}}, \text{frame})$ with the *same signature* as the mouse/rat/macaque loaders, so any FDTD or angular-spectrum driver wired against one species runs on this one unchanged. The slab ramp is the flagged placeholder (§4); the cavity is overwritten with brain-parenchymal properties ($c = 1540 \text{ m/s}$, $\rho = 1040 \text{ kg/m}^3$) exactly as in the other pillars. This slab is the hand-off boundary to the wave-solver siblings.

7 Out of scope here: the wave-physics study (deliverables 3–7)

TUBA is the geometry/atlas/targeting layer; the wave physics lives in the `fullwave2-*` FDTD siblings and consumes the (c, ρ, dz) slab and placement this pillar exports. The funded study’s remaining deliverables are therefore deliberately *not* in this branch:

- D3 time-reversal aberration correction through the individual skull (virtual source at each target → recorded element signals → phase-conjugate transmit delays);
- D4 multifocal beam synthesis — two simultaneous foci (S1 + M1) with independent amplitude control, plus at least one alternative to superposed time reversal (e.g. Gerchberg–Saxton);
- D5 the feasibility study (highest-priority result): sweep two-foci separation 1.5–4.0 mm at 2, 3, 4 MHz through the skull model, reporting –6 dB focal dimensions, peak-to-valley contrast, off-target ratio, and transcranial focal gain;
- D6 safety margins — MI at target and skull, and a bioheat estimate over the 400 kPa / 300 ms / 3 s duty cycle;
- D7 the per-animal targeting artifact (element delays, apodisations, target metadata, safety numbers).

The PPW budget (Table 2) and the time-of-flight grid-convergence (§4) are the model-side inputs D3 and D5 will build on.

8 Verification

Because no scan is staged, verification is by offline unit tests (`tests/test_saimiri.py`, 11 tests, no network / no antspyx / no data), covering the logic that must hold regardless:

- importing the pillar stays ANTs-free (ANTs is lazy inside `tuba.core.warp` and the atlas binding);
- the calibration guard refuses uncalibrated input on every entry point, and the calibrated Aubry mapping is physically monotonic (porosity ↓, $\rho/c/\alpha$ ↑ with HU) with the correct water / cortical endpoints and clamping;
- the placeholder ramp is flagged and maps to nominal endpoints;
- the PPW, time-of-flight-aberration, and grid-convergence maths (homogeneous ⇒ 0; bone slab ⇒ analytic; smooth profile converges);
- VALiDATE29 file/label resolution and S1/M1 target keyword mapping on a synthetic atlas, plus a synthetic affine registration round-trip;
- the data manifest carries the three Saimiri entries with the expected calibration/license semantics.

The full repository suite is 42 passing (31 prior + 11 new); `ruff` is clean.

9 Open items

The pillar's machinery is complete and tested; the remaining items all require the staged scan (and, for quantitative acoustics, an HU-calibrated colony CT):

1. Stage the museum microCT (`python -m tuba.data.fetch_saimiri`) and the VALiDATE29 atlas (`python -m tuba.data.fetch_validate29`), then run `python -m tuba.species.saimiri build`.
2. Calibrate the PROVISIONAL constants (orientation flips, intensity thresholds, cavity hull) against the staged scan via the orientation/histogram probe, exactly as the rat pillar [10] documents; report the measured cavity volume against the 20–28 mL target and produce the cavity + registration QC figures.
3. Confirm the VALiDATE29 cortical-label spellings against the staged LUT and pin them (§3); add the stereotaxic hand-sub-representation prior.
4. When a Vanderbilt colony CT is available, set CT_CALIBRATION to an hu calibration to unlock the quantitative Aubry mapping, and run the grid-convergence with the real property field.
5. Hand the exported slab + targets to the fullwave2-* siblings for deliverables 3–7 (§7).

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